ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 7

Implement SVM Algorithm for Regression

Dataset: Scikit-learn Diabetes Dataset

1. Aim

To implement Support Vector Regression (SVR) for predicting the continuous Diabetes target and study the effect of the regularization parameter C on regression performance.

2. Objectives

- Understand the concept of Support Vector Regression.
- Use an RBF kernel to model nonlinear relationships.
- Standardize input features before SVR.
- Evaluate the model using MAE, MSE, RMSE and R^2.
- Compare different C values.

3. Dataset Description

The Diabetes dataset contains 442 observations, 10 independent features and a continuous target. The supplied SVR experiment uses an 80:20 train-test split, with 353 training and 89 testing observations.

Property	Value
Samples	442
Features	10
Kernel	RBF
C	100
Gamma	scale
Epsilon	0.1

4. Theory

Support Vector Regression is the regression counterpart of Support Vector Machines. Instead of minimizing error for every observation in the same way as ordinary least squares, SVR seeks a function that fits the data while controlling model complexity and allowing a tolerance region around predictions.

SVR introduces an epsilon-insensitive loss. Errors within an epsilon-wide tube around the regression function are not penalized. Observations outside the tube contribute to the optimization and may become support vectors.

The parameter C controls the penalty assigned to observations outside the epsilon tube. A larger C generally allows the model to focus more strongly on reducing training errors, while a smaller C imposes stronger regularization.

The kernel function allows SVR to represent nonlinear relationships. The RBF (Radial Basis Function) kernel measures similarity based on distance and can create nonlinear decision surfaces in the transformed feature space.

The RBF kernel is commonly written as $K(x,x') = \exp(-\gamma||x-x'||^2)$. The gamma parameter controls how rapidly the influence of a training observation decreases with distance.

Feature scaling is important for SVR because the kernel and distance calculations are sensitive to feature magnitudes. Standardization makes the input features comparable before fitting the model.

MAE, MSE and RMSE quantify prediction error, while R^2 measures the proportion of variance explained by the model. Using several metrics gives a more complete picture of performance.

Hyperparameters such as C, gamma and epsilon can strongly affect SVR performance. They are normally selected using validation or cross-validation rather than relying only on the final test set.

The SVR optimization balances two objectives: keeping the regression function sufficiently flat and allowing a controlled number of observations to lie outside the epsilon-insensitive tube. This balance helps SVR generalize beyond the training data.

Support vectors are the observations that determine the final regression function. Points comfortably inside the epsilon tube do not directly influence the solution in the same way as points on or outside the tube.

In this experiment, an RBF kernel is used with gamma='scale' and epsilon = 0.1. The supplied analysis tests C values of 0.1, 1, 10, 100 and 1000. The best recorded R^2 occurs at C = 100.

5. Methodology

- Load the Diabetes dataset.
- Split the observations into training and testing sets.
- Standardize the ten input features.
- Create an SVR model using an RBF kernel, C = 100, gamma = scale and epsilon = 0.1.
- Evaluate the model using MAE, MSE, RMSE and R^2.
- Train/evaluate the supplied C values and compare R^2 and RMSE.
- Visualize actual versus predicted values and residuals.

6. Outputs

6.1 SVR Model Metrics

Metric	Value
Mean Absolute Error (MAE)	39.4187
Mean Squared Error (MSE)	2607.5423
Root Mean Squared Error (RMSE)	51.0641
R2 Score	0.5078

6.2 C Parameter Comparison

C	R2 Score	RMSE
0.1000	0.0227	71.9593
1.0000	0.1822	65.8235
10.0000	0.4937	51.7918
100.0000	0.5078	51.0641
1000.0000	0.1796	65.9278

6.3 Actual vs Predicted

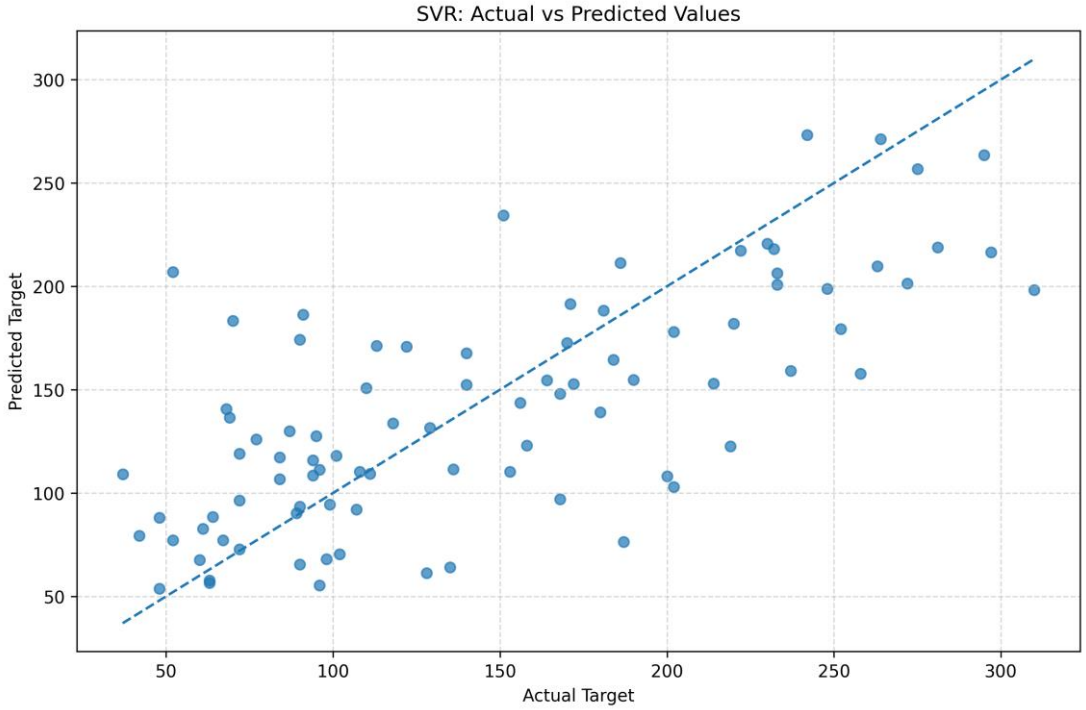


Figure 1: SVR actual versus predicted values

6.4 Residual Plot

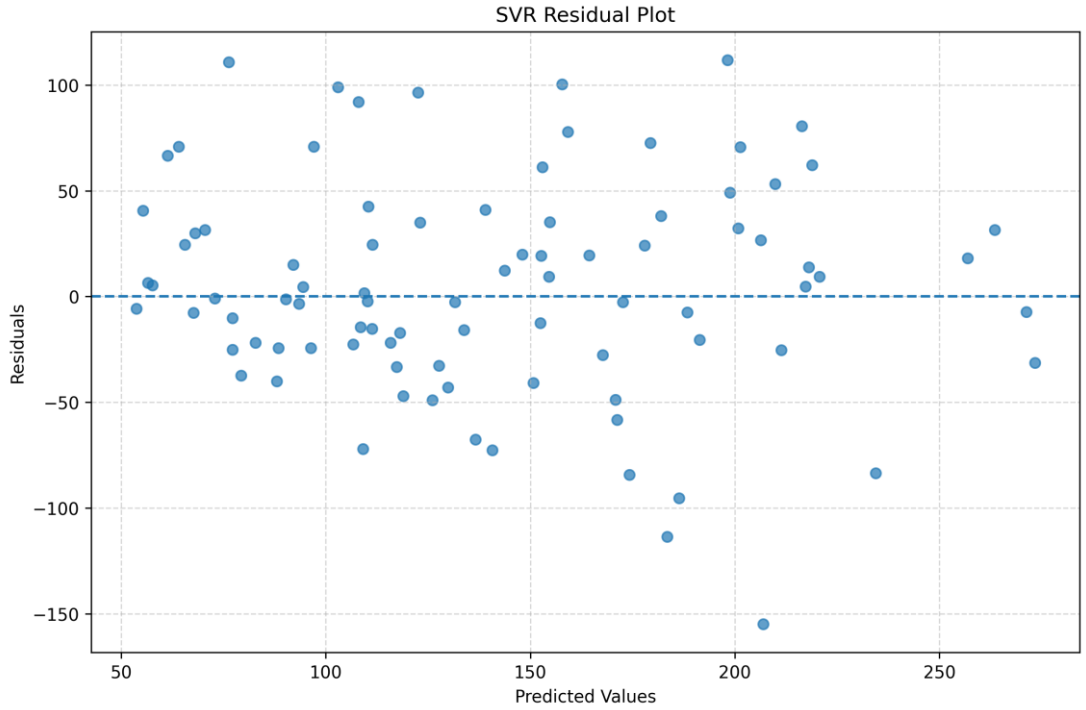


Figure 2: SVR residual plot

6.5 Test Actual vs Predicted

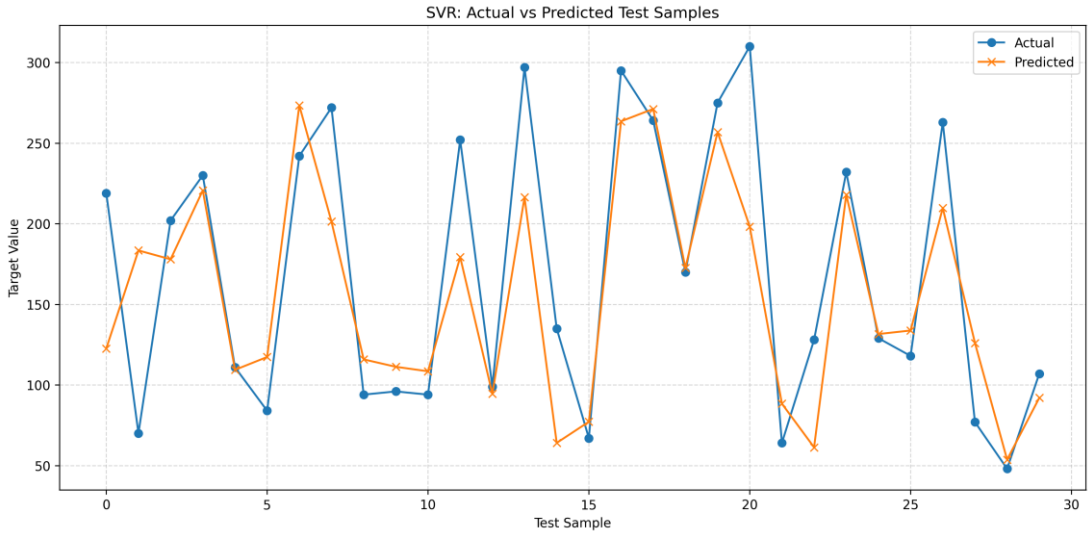


Figure 3: Actual and predicted test samples

6.6 C vs R²

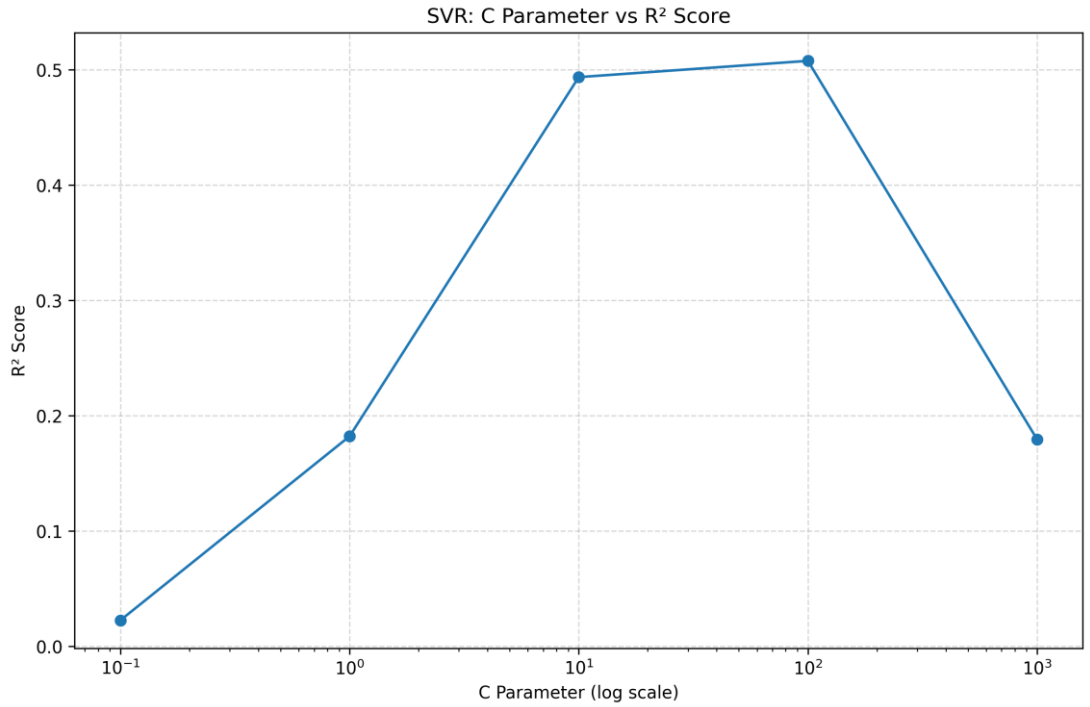


Figure 4: Effect of C on R² score

7. Result

The SVR model achieved MAE = 39.4187, MSE = 2607.5423, RMSE = 51.0641 and R^2 = 0.5078. Among the tested C values, C = 100 produced the highest recorded R^2 of 0.5078.

8. Observations

- The RBF kernel allows nonlinear relationships to be modeled.
- Feature scaling was applied because SVR is sensitive to feature magnitudes.
- C = 100 performed better than the tested smaller and larger values in the supplied comparison.
- The model achieved a higher R^2 than the supplied Linear Regression result.
- Residual and actual-versus-predicted plots help assess prediction behavior.

9. Conclusion

SVR was successfully implemented using an RBF kernel. The experiment demonstrated the importance of feature scaling and hyperparameter selection. The supplied comparison identified C = 100 as the best tested value and produced an R^2 of 0.5078.